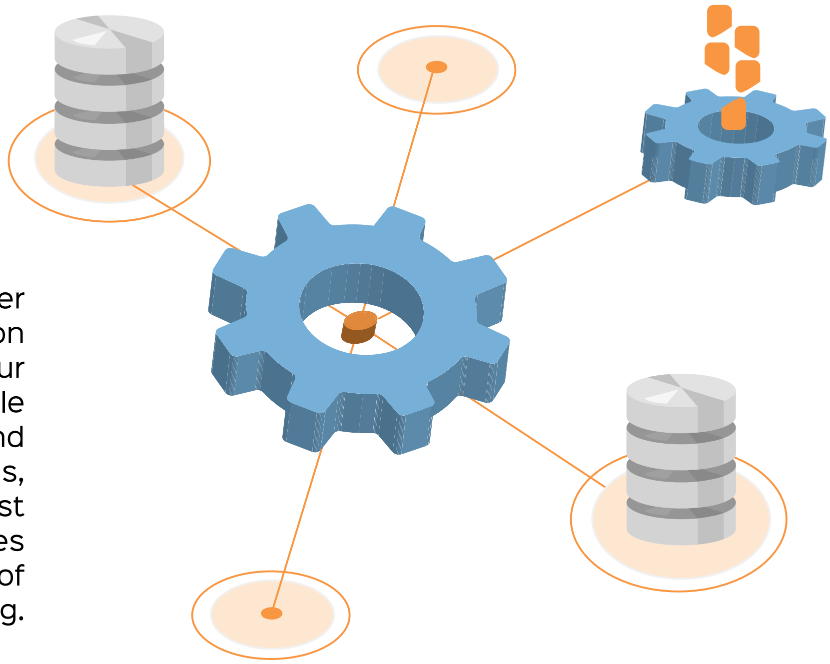
Demystifying Connection Pools: A Deep Dive



By understanding the inner workings of connection pooling and building your own pools, you will be able to build more reliable and high-performing applications, especially at scale. This first part of a two-part series demystifies the magic of connection pooling.

If you're a software engineer, you've likely encountered the need to manage database connections efficiently. That's where connection pooling comes in! Connection pools are a critical aspect of software engineering. They allow applications to efficiently manage connections to a database or any other system. If your application requires constant access to a system, establishing a new connection to the system for every request can quickly become resource-intensive, causing the application to slow down or even crash. This is where connection pools come in.

As engineers, we often don't spend a lot of time thinking about connections. A single connection is typically inexpensive. However, as things scale up, the costs of creating and maintaining these connections increase accordingly. This is why I believe understanding the world of connection pooling is important. It will enable us to build more performant and reliable applications, especially at scale.

Typical connections

Before jumping to connection pooling, let us understand how an application typically connects to a system to perform any operation:

1. The application attempts to open a connection.
2. A network socket is opened to connect the application to the system.
3. Authentication is performed.
4. The operation is performed.
5. The connection is closed.
6. The socket is closed.

As you can see, opening and closing the connection and the network socket is a multi-step process that requires resource computation. However, not closing the connection, or keeping it idle, also consumes resources. This is why we need connection pooling. While you will mostly see connection pooling being used in database systems, the concept can be extended to any application that communicates with a remote system over a network.

What are connection pools?

Connection pools are typically a cache of connections that can be reused by an application. Instead of creating a new connection each time an application needs to interact with the system, a connection is borrowed from the pool. When it's no longer needed, it is returned to the pool to be reused later.